

# Package ‘PBGof’

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**Type** Package

**Title** Parametric Bootstrap Tests for the Skew-Normal Distribution

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**Depends** R (>= 4.1)

**Description** Provides goodness-of-fit tests for the skew-normal distribution with estimated parameters. Implements Kolmogorov-Smirnov and Cramér-von Mises tests using parametric bootstrap or precomputed simulation quantiles, together with robust parameter estimation procedures. Package methods and documentation are described by Li and Khang (2026)  
<<https://github.com/Divo-Lee/PBGof>>.

**Imports** methods, sn

**Suggests** knitr, rmarkdown

**VignetteBuilder** knitr

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PBGof-precomputed-tests

*Fast Skew-Normal Goodness-of-Fit Tests Using Precomputed Quantiles***Description**

Computes a skew-normal goodness-of-fit statistic and obtains an approximate p-value from a pre-computed 100,000-replicate quantile table.

**Usage**

```
PBGof_ks_test(data, ks_table = NULL)
```

```
PBGof_cvm_test(data, cvm_table = NULL)
```

**Arguments**

`data` A numeric vector whose length is represented in the table.

`ks_table, cvm_table` Optional custom quantile tables. If NULL, the corresponding table bundled with PBGoF is used.

**Details**

The fitted absolute CP skewness is rounded to two decimal places and truncated to  $[0.01, 0.99]$ . P-values are conservative step-function approximations between 0.01 and 0.99. No interpolation is performed.

If the fitted `gamma1` is negative, table lookup uses `abs(gamma1)`. Reflection of a skew-normal variable changes the signs of the DP shape parameter and CP skewness, but the null distributions of the EDF statistics are invariant to this sign change (Mateu-Figueras et al., 2007). The bundled tables therefore require only non-negative skewness values.

For samples larger than 500, all observations remain in the fit and empirical distribution function, while the external statistic multiplier and table row use `n_used = 500`.

**Value**

A list containing `statistic`, actual sample size `n`, lookup and scaling sample size `n_used`, `gamma1_hat`, `gamma1_used`, and `p.value`.

**References**

Mateu-Figueras, G., Puig, P., and Pewsey, A. (2007). Goodness-of-fit tests for the skew-normal distribution when the parameters are estimated from the data. *Communications in Statistics—Theory and Methods*, 36(9), 1735–1755. doi:[10.1080/03610920601126217](https://doi.org/10.1080/03610920601126217)

**Examples**

```
set.seed(123)
x <- sn::rsn(50, xi = 0, omega = 1, alpha = 3)
PBGof_ks_test(x)
PBGof_cvm_test(x)
```

---

sn-parametric-bootstrap

*Parametric Bootstrap Goodness-of-Fit Tests for a Skew-Normal Distribution*

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**Description**

Tests skew-normal goodness of fit using a parametric bootstrap, re-estimating the model in each bootstrap sample with `sn.fit.robust()`.

**Usage**

```
sn.param.bootstrap.ks.test(data = NULL, B = 1000L, seed = 103L,
  verbose = FALSE)

sn.param.bootstrap.cvm.test(data = NULL, B = 1000L, seed = 103L,
  verbose = FALSE)
```

**Arguments**

|         |                                                                                                                          |
|---------|--------------------------------------------------------------------------------------------------------------------------|
| data    | A numeric vector containing at least 10 finite observations.                                                             |
| B       | A positive integer giving the number of bootstrap samples.                                                               |
| seed    | A single integer, or NULL to use the current random-number stream. The caller's random-number state is restored on exit. |
| verbose | Logical; whether to print a short summary.                                                                               |

**Details**

The KS function uses  $\sqrt{n}D$ ; the CvM function uses the Cramér–von Mises statistic. Failed fits are excluded and the p-value uses the finite-simulation correction  $(1 + \sum I(T_b \geq T_0)) / (B_{\text{valid}} + 1)$ . The names ending in .1 are compatibility aliases.

**Value**

A single numeric bootstrap p-value with attributes `statistic`, `B`, `valid`, and `failed`.

## Examples

```
set.seed(123)
x <- sn::rsn(50, xi = 0, omega = 1, alpha = 3)
sn.para.bootstrap.ks.test(x, B = 99)
sn.para.bootstrap.cvm.test(x, B = 99)
```

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sn.fit.robust

*Numerically Robust Fitting of the Skew-Normal Distribution*


---

## Description

Fits a skew-normal distribution by MLE, then falls back to default penalized MLE and matching-prior penalized MLE when necessary.

## Usage

```
sn.fit.robust(data = NULL, para_form = c("DP", "CP"))
```

## Arguments

|           |                                                                                             |
|-----------|---------------------------------------------------------------------------------------------|
| data      | A numeric vector containing at least 10 finite observations.                                |
| para_form | Parameterization of the result: "DP" for direct parameters or "CP" for centered parameters. |

## Details

Robustness here concerns numerical fitting failures, not resistance to outliers or model contamination.

## Value

A named numeric vector containing parameter estimates and standard errors. If all fitting procedures fail, all entries are NA.

## Examples

```
set.seed(123)
x <- sn::rsn(100, xi = 0, omega = 1, alpha = 5)
sn.fit.robust(x, "DP")
sn.fit.robust(x, "CP")
```

sn.plot.check

*Graphical Check of a Fitted Skew-Normal Distribution***Description**

Draws a density-scale histogram of the observed data and overlays a skew-normal density fitted by `sn.fit.robust()`.

**Usage**

```
sn.plot.check(
  data,
  breaks = "FD",
  col = "grey85",
  border = "white",
  curve_col = "#6CC6C6",
  lwd = 2,
  main = NULL,
  xlab = "Values",
  add_rug = TRUE,
  ...
)
```

**Arguments**

|                        |                                                                          |
|------------------------|--------------------------------------------------------------------------|
| <code>data</code>      | A numeric vector containing at least 10 finite observations.             |
| <code>breaks</code>    | Histogram breaks passed to <code>graphics::hist()</code> .               |
| <code>col</code>       | Fill color for the histogram.                                            |
| <code>border</code>    | Border color for histogram bars.                                         |
| <code>curve_col</code> | Color of the fitted skew-normal density curve.                           |
| <code>lwd</code>       | Line width of the fitted density curve.                                  |
| <code>main</code>      | Main plot title.                                                         |
| <code>xlab</code>      | Label for the horizontal axis.                                           |
| <code>add_rug</code>   | Logical; whether to add observations as a rug plot.                      |
| <code>...</code>       | Additional graphical arguments passed to <code>graphics::hist()</code> . |

**Details**

This visual model check complements, but does not replace, the formal goodness-of-fit tests provided by `PBGoF`.

**Value**

Invisibly returns a list containing the fitted DP parameters, histogram object, and coordinates of the fitted density curve.

**Examples**

```
set.seed(123)
x <- sn::rsn(100, xi = 0, omega = 1, alpha = 4)
sn.plot.check(x)
```

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